

LAC EON, QC, Canada

WMO: 714210

Lat: 51.867N

Lon: 63.283W

Elev: 589

StdP: 94.45

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.4	-31.0	-37.7	0.1	-33.4	-35.2	0.1	-30.7	9.5	-11.8	8.8	-12.9	2.3	320	0.740

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	25.6	17.1	23.3	15.6	21.3	14.7	18.9	23.0	17.2	21.0	16.0	19.6	3.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.5	13.4	20.6	15.8	12.1	18.8	14.6	11.1	17.5	56.3	22.7	50.9	20.8	46.9	19.6	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.3	6.5	DB	-38.3	29.2	2.6	3.1	-40.1	31.4	-41.7	33.2	-43.1	35.0	-45.0	37.2	(4)
(5)			WB	-38.6	20.2	2.7	2.0	-40.5	21.7	-42.0	22.8	-43.5	23.9	-45.5	25.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.4	-18.9	-17.7	-11.4	-3.3	3.9	10.6	14.9	14.1	8.4	1.7	-5.9	-14.2	(6)	
	DBStd	12.85	7.26	6.90	6.86	5.10	3.98	4.51	3.61	3.52	3.89	4.10	5.54	6.78	(7)	
	HDD10.0	4544	897	775	663	399	194	44	3	6	77	259	478	749	(8)	
	HDD18.3	7222	1156	1008	921	649	447	235	117	139	299	516	727	1008	(9)	
	CDD10.0	386	0	0	0	0	5	63	157	132	28	2	0	0	(10)	
	CDD18.3	23	0	0	0	0	0	4	11	8	0	0	0	0	(11)	
	CDH23.3	204	0	0	0	0	2	43	102	57	1	0	0	0	(12)	
	CDH26.7	46	0	0	0	0	0	10	27	8	0	0	0	0	(13)	
(14) Wind	WSAvg	3.2	3.1	3.2	3.5	3.5	3.1	3.1	3.0	2.9	3.2	3.3	3.3	3.1	(14)	
(15) Precipitation	PrecAvg	1058	70	64	82	61	78	87	113	116	91	108	97	91	(15)	
	PrecMax	1271	177	119	182	95	168	131	203	198	154	257	149	167	(16)	
	PrecMin	790	21	9	24	10	32	48	50	36	48	45	43	18	(17)	
	PrecStd	148	38	27	41	25	37	25	40	38	29	50	30	37	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.9	2.2	6.7	13.1	22.0	27.7	29.7	27.8	21.9	16.2	11.7	1.8	(19)	
		MCWB	0.3	1.1	2.3	5.8	12.4	16.6	21.6	19.5	14.8	11.6	10.4	1.5	(20)	
	2%	DB	-1.4	-0.3	2.8	8.8	18.2	23.7	25.9	24.8	19.2	12.3	6.2	0.1	(21)	
		MCWB	-1.7	-1.0	0.0	2.9	9.6	14.7	17.2	17.6	13.7	9.4	5.2	-0.1	(22)	
	5%	DB	-4.0	-2.9	0.6	6.3	14.7	21.0	23.6	22.6	17.0	10.2	3.5	-1.2	(23)	
		MCWB	-4.4	-3.7	-1.2	2.0	7.8	13.2	15.8	16.1	13.1	7.8	2.6	-1.6	(24)	
	10%	DB	-7.2	-6.2	-1.0	4.2	11.9	18.7	21.5	20.4	14.9	8.1	1.3	-3.8	(25)	
		MCWB	-7.8	-7.1	-2.8	0.9	6.6	12.2	15.0	15.1	11.9	6.2	0.4	-4.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.1	0.9	3.3	6.7	13.1	19.0	23.2	22.6	17.3	12.2	11.3	1.7	(27)	
		MCDB	0.3	1.2	6.1	10.4	19.5	24.3	29.8	25.3	19.5	14.2	11.5	1.8	(28)	
	2%	WB	-1.9	-1.5	0.7	4.3	10.7	16.5	19.2	18.7	15.0	9.8	5.6	-0.1	(29)	
		MCDB	-1.6	-0.9	2.2	7.6	16.7	20.8	23.2	22.5	16.7	11.4	6.2	0.1	(30)	
	5%	WB	-4.5	-4.2	-1.0	2.6	8.8	15.0	17.4	17.1	13.4	8.3	3.0	-1.6	(31)	
		MCDB	-4.1	-3.3	0.7	5.5	13.7	19.5	21.5	20.6	16.2	9.8	3.6	-1.2	(32)	
	10%	WB	-7.8	-7.4	-2.8	1.3	7.1	13.3	16.1	15.7	11.8	6.3	0.6	-4.4	(33)	
		MCDB	-7.3	-6.5	-1.1	3.8	11.7	17.3	19.9	19.0	14.5	7.7	1.2	-3.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.5	11.5	12.3	10.4	10.2	10.4	9.7	9.2	8.1	6.3	7.0	8.7	(35)	
		MCDBR	10.7	11.1	12.6	14.0	17.1	15.0	13.1	12.7	11.5	10.1	8.3	7.9	(36)	
		MCWBR	10.0	10.5	10.7	9.4	9.7	7.1	6.0	6.6	7.1	7.1	7.9	8.1	(37)	
	5% WB	MCDBR	10.1	10.8	10.9	11.6	15.0	12.4	10.8	11.0	9.8	8.5	8.2	7.9	(38)	
		MCWBR	10.2	10.0	9.4	8.3	9.2	7.0	6.1	6.9	7.3	6.9	8.2	8.1	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.235	0.245	0.263	0.318	0.350	0.353	0.387	0.352	0.328	0.294	0.279	0.241	(40)	
	taud		2.303	2.293	2.274	2.190	2.373	2.432	2.366	2.471	2.508	2.571	2.400	2.288	(41)	
	Ebn at Noon		801	896	943	917	903	901	860	878	863	835	734	720	(42)	
	Edh at Noon		61	82	103	129	114	108	114	98	84	64	57	52	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.09	2.14	3.48	4.70	5.15	5.14	4.80	4.00	3.02	1.78	0.93	0.70	(44)	
	RadStd		0.15	0.17	0.39	0.45	0.44	0.48	0.39	0.22	0.24	0.13	0.07	0.09	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	+1.22	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page